

Delta Engine (MKM-DE-001)

Wolfram Oracle Benchmark of FS01, the Risky Annuity and Mark-to-Market

MKM Research Labs

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1 Executive Summary

This document benchmarks the Delta Engine (MKM-DE-001), which values and risk-manages the PRS book: for every trade it computes the risky annuity, the flood sensitivity FS01 and the mark-to-market. It runs in production, behind the trading routes, the blotter, the XVA service and the historical end-of-day series. The fair spread it differences is PR-001's, benchmarked separately.

The code computes its documented formulae, to the precision it reports. Over 120 cases — annual hazard rates 0.0001–0.5, tenors 1 to 10 years, a flat 3% curve and the book's yield curve, notional GBP 10,000,000 — the risky annuity agrees with a 50-digit reference to 5.8×10^{-16}, and the spread change, FS01, the returned annuity, the hazard sensitivity and mark-to-market agree to within half a unit of the rounding the engine applies. Wolfram and the 50-digit reference agree to 4.9×10^{-25}, so the reference is confirmed by a second, independent implementation.

What changed since v0.1. Benchmark v0.1 tested the engine against model document v1.0 and found two defects in the code and three errors in the document:

1. **MTM used a rounded annuity**, rounded to six places for display, which moved MTM by up to GBP 3.40 on GBP 10m. The engine now values with the unrounded annuity; MTM now agrees to the penny.
2. **FS01 was silent on the spread floor.** At $h = 0.0001$ the fair spread sits on the 2 bp floor and FS01 reflects a 0.0078 bp spread change — the right P&L sensitivity, since MTM does not move either, but blind to the flood risk. The engine now flags the floor and reports the hazard sensitivity beside FS01: 1.0039 bp, against an exact 1.0039 bp.
3. **The document** said FS01 scaled linearly with the hazard rate (it is nearly flat), put the yield sensitivity at 3–5% (it is 0.4–2.4%), and gave the worked annuity as 2.738 (it is 2.724084). Model document v1.1 corrects all three, states that the basis delta *is* the gauge delta, and documents the floor.

The model document's claims (v1.1, Table 2): all 7 of the 7 claims tested hold, each to the precision the document quotes: FS01 at 500 bp is 0.94 times FS01 at 50 bp; a 100 bp yield rise lowers FS01 by 0.4–2.4%; a 5-year trade carries 4.53 times the FS01 of a 1-year; the worked annuity is 2.724084; and the 1 bp bump is within 3.0×10^{-4} bp of the exact gradient. A 50 bp move is 0.32% from full revaluation.

Evidence. 8 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `81c145f8f330`:
  `src/models/trading/delta_engine/pricer.py`,
  `src/models/trading/delta_engine/engine.py`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-26 by `python -m tools.wolfram_oracle de001`.

2 Model Purpose and Scope

2.1 What is benchmarked

`compute_risky_annuity`, `compute_gauge_delta` (FS01, the floor flag and the hazard sensitivity), `compute_basis_delta` and `compute_mark_to_market` in `src/models/trading/delta_engine/pricer.py`, called as `DeltaEngine.enrich_trade` calls them, with the rounding they apply, against model document v1.1 (`delta_engine_v1.1.pdf`).

2.2 What is not

The fair spread itself (MKM-PR-001's benchmark); the market-state inputs the trader maintains; the risk-grid aggregation, which is a sum. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

With annual hazard rate h, $\lambda = -\ln(1-h)$, quarterly dates $t_i = i/4$ and discount $D(t) = e^{-r(t)t}$, the document defines

$$A(h,T) = \sum_{i=1}^{4T} \tfrac{1}{4}\, e^{-\lambda t_i} D(t_i), \qquad \mathrm{FS01} = \delta\, \frac{s(h+10^{-4}) - s(h)}{10^4}\, A(h,T)\, N, \qquad \mathrm{MTM} = \delta\, \frac{s - s_0}{10^4}\, A\, N, \tag{1}$$

with s PR-001's fair spread, floored at 2 bp, $\delta = \pm 1$ for a payer or receiver, and N the notional.

Why FS01 is flat in h. On a flat curve PR-001's spread has the closed form $s = 10^4(1-R)\,(e^{\lambda\Delta t} - 1)\, e^{r\Delta t/2}/\Delta t$, so

$$\frac{\mathrm{d}s}{\mathrm{d}h} = 10^4\,(1-R)\,\frac{e^{\lambda\Delta t}\, e^{r\Delta t/2}}{1-h} \approx 10^4. \tag{2}$$

A 1 bp bump in h moves the spread by about 1 bp at any level, and FS01 is about $10^{-4} A\, N$: it follows the annuity, which falls slowly as h rises.

4 Benchmark Method

The equations above are evaluated with `mpmath` at 50 significant digits, from the documents rather than the code; the exact gradient $\mathrm{d}s/\mathrm{d}h$ is a high-precision central difference, checked against the closed form above. Wolfram evaluates the same sums exactly, with the book curve as a piecewise function, differentiates the spread symbolically, and evaluates the bumped spread at the very double the engine computes, $h + 10^{-4}$ in floating point. The engine rounds what it returns, so each rounded output is judged to half a unit in its last place. MTM uses an inception spread of 150 bp.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Risky annuity, unrounded (s3.1)	5.8×10^{-16}	1.0×10^{-12}	pass
Spread change Δs (bp, to 10^{-4}): model vs reference	5.0×10^{-5}	5.0×10^{-5}	pass
FS01 (GBP, to the penny): model vs reference	5.0×10^{-3}	5.0×10^{-3}	pass
Risky annuity as returned (to 10^{-6}): model vs reference	5.0×10^{-7}	5.0×10^{-7}	pass
Mark-to-market, unrounded annuity (GBP, to the penny): model vs reference	5.0×10^{-3}	5.0×10^{-3}	pass
Hazard sensitivity, unfloored spread (bp, to 10^{-4}): model vs reference	5.0×10^{-5}	5.0×10^{-5}	pass
Wolfram agrees with the reference: annuity, Δs, gradient	4.9×10^{-25}	1.0×10^{-12}	pass
Wolfram agrees with the reference: claims (annuity, FS01)	1.4×10^{-31}	1.0×10^{-12}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Document says	Benchmark finds	Outcome
s6 item 5: 1 bp bump error	$<$ 0.01 bp vs analytical gradient	3.0×10^{-4} bp at most	met
s6 item 1: FS01 against h	FS01(500)/FS01(50) = 0.94	0.9436	met
s6 item 2: tenor	5Y $\approx 4.5\times$ 1Y	4.53×	met
s6 item 3: yields +100 bp	FS01 falls 0.4–2.4%	FS01 falls 0.44–2.40%	met
s7.2 hand calculation	$A = 2.724084$	$A = 2.72408398$	met
s8 item 1: 50 bp move	FS01×Δh differs from revaluation	by 0.32%	finding
s3.4 spread floor	flagged; hazard sensitivity reported	flag right in every case; at $h = 0.0001$: FS01 $\Delta s = 0.0078$, hazard 1.0039 vs gradient 1.0039 bp	met
s3.5 basis delta	is the gauge delta	identical in every case	met

5.2 FS01 across the grid

Table 3: FS01 in GBP from the engine, flat 3% curve, $R = 0$, notional 10,000,000. Down a column it barely moves: FS01 is set by the annuity, not the hazard rate

h	1Y	2Y	3Y	4Y	5Y	10Y
0.0001	8	15	22	29	36	67
0.0005	986	1,941	2,869	3,768	4,640	8,624
0.001	986	1,942	2,868	3,766	4,637	8,609
0.005	988	1,943	2,864	3,754	4,613	8,485
0.01	991	1,944	2,859	3,738	4,583	8,334
0.02	998	1,947	2,849	3,707	4,524	8,043
0.05	1,018	1,956	2,820	3,618	4,353	7,252
0.1	1,053	1,973	2,776	3,477	4,090	6,169
0.2	1,135	2,017	2,701	3,233	3,645	4,673
0.5	1,551	2,303	2,668	2,845	2,931	3,010

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The engine is shown to compute its documented formulae; whether a parallel 1 bp bump in the annual rate is the right risk measure is a separate question.

2. **The floor.** The engine now flags it, but risk limits and the risk grid aggregate FS01 alone; positions on the floor contribute almost nothing to them. Whether limits should also read the hazard sensitivity is for the desk and the model owner.

3. **Parallel bump only.** FS01 bumps the annual rate in parallel across tenors; term-structure and cross-gauge sensitivities are outside this benchmark.

7 Document History

Date	Version	Description	Author
26-Oct-2026	0.1	First issue, against model document v1.0	David K Kelly
26-Oct-2026	0.2	After the MTM and floor fixes, against model document v1.1	David K Kelly